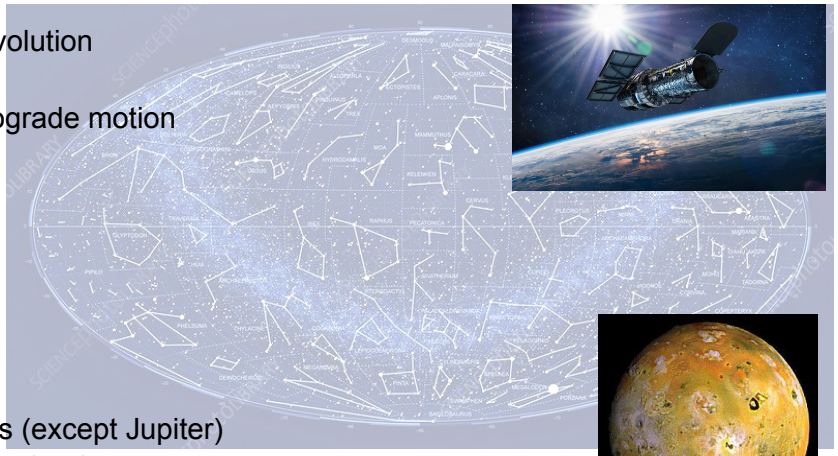
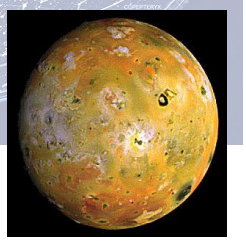
Advanced Astronomy: Solar System

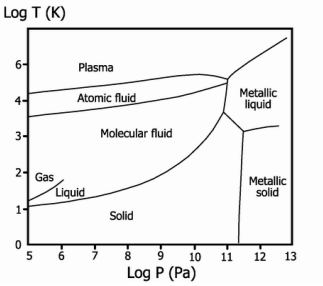


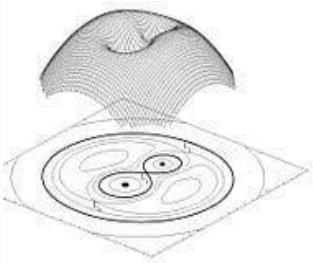


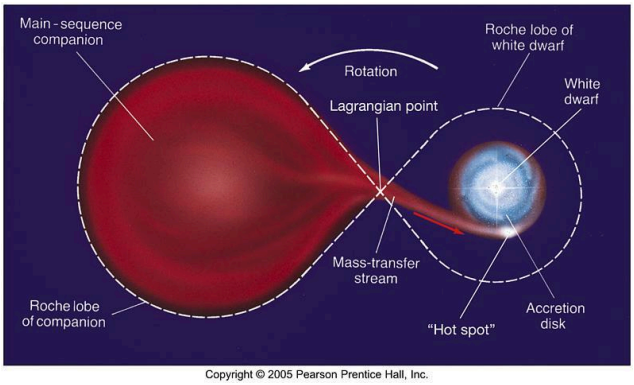
- Aristarchus and the Copernican Revolution
  - Solstices and equinoxes
  - Star charts, the ecliptic, retrograde motion
  - Heliocentrism
- Telescopes
- Astrodynamics
- Formation of the solar system
- Astrophysics of the Sun
- Astrophysics of Jupiter
- Projects on (time permitting)
  - Terrestrial and Jovian planets (except Jupiter)
  - Asteroid belt/Kuiper belt/Oort cloud

In the process, we will learn astrophysics concepts and how to apply them including

- Astro-geometry
  - Parallax
  - Angular sizes and distances
  - Ellipses and eccentricity
- Newtonian astrophysics
  - Kepler's laws
  - Mass determination and barycenter
  - Orbital and escape speeds
  - Atmospheres
  - Rotating frames of reference, Lagrange points, and orbital dynamics
  - Roche limits and fragmentation
- Nuclear and Particle physics I
  - Simplified Model of Particle Physics
  - Mass-energy equivalence
  - Nuclear reactions







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